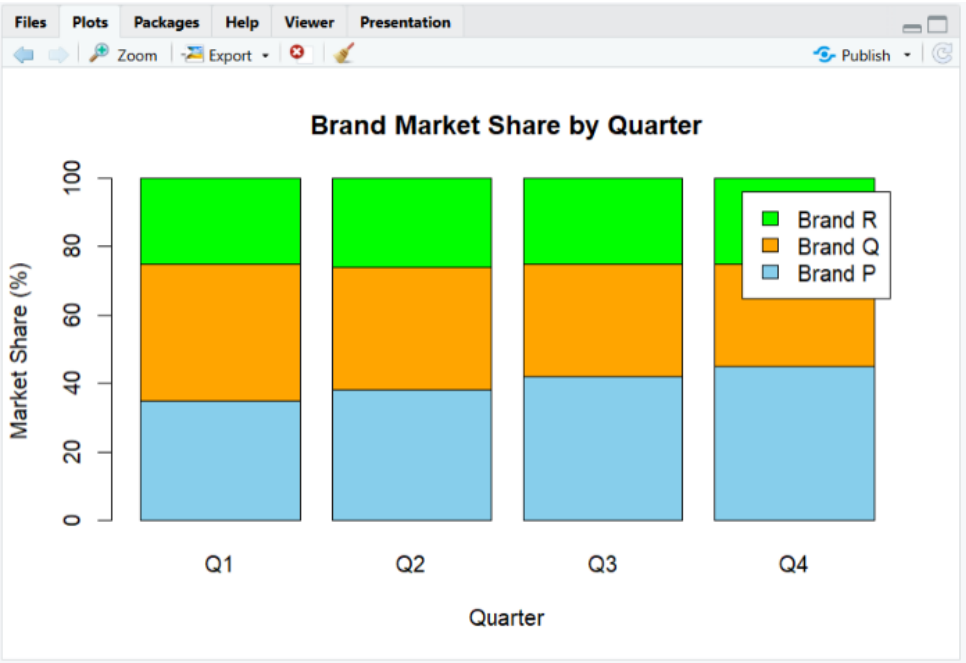
Q1. Stacked Bar Charts and Proportions — R

1. Stacked bar chart

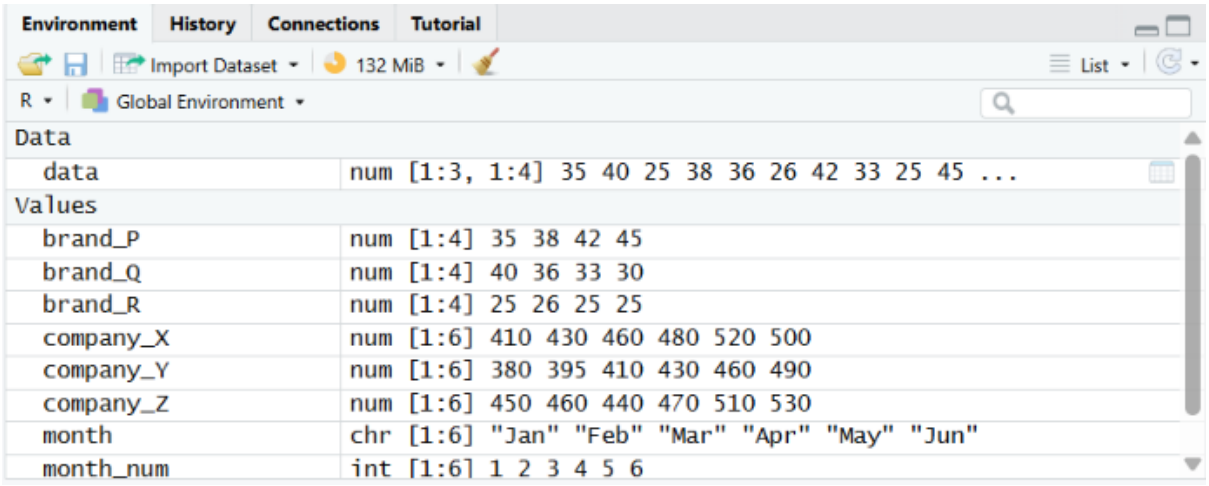
```
quarter <- c("Q1", "Q2", "Q3", "Q4")
brand_P <- c(35, 38, 42, 45)
brand_Q <- c(40, 36, 33, 30)
brand_R <- c(25, 26, 25, 25)
data <- rbind(brand_P, brand_Q, brand_R)
barplot(data,
        names.arg = quarter,
        col = c("skyblue", "orange", "green"),
        main = "Brand Market Share by Quarter",
        xlab = "Quarter",
        ylab = "Market Share (%)",
        legend.text = c("Brand P", "Brand Q", "Brand R"))
```



2. 100% proportional stacked bar chart

Since every quarter already totals **100%**, the given data can be directly displayed as a proportional stacked bar chart.

```
barplot(data,
        names.arg = quarter,
        col = c("skyblue", "orange", "green"),
        main = "100% Proportional Market Share",
        xlab = "Quarter",
        ylab = "Percentage (%)",
        legend.text = c("Brand P", "Brand Q", "Brand R"),
        ylim = c(0, 100))
```



3. Comparison

- **Stacked bar chart:** Shows the individual market shares and the total market share for each quarter.
- **100% proportional chart:** Makes it easier to compare the **percentage contribution** of each brand within each quarter.
- Since each quarter totals 100%, the proportional chart clearly shows how the market is divided among P, Q and R.

4. Brand with consistently rising trend

Brand P has a consistently rising trend:

35% → 38% → 42% → 45%

5. Limitation

A limitation of stacked bar charts is that **individual segments are difficult to compare**, especially segments that do not start from the same baseline. The middle and upper segments are harder to visually compare accurately.

Q2. Legends and Annotations — R

1. Monthly stock-price trend

```
month <- c("Jan", "Feb", "Mar", "Apr", "May", "Jun")
company_X <- c(410, 430, 460, 480, 520, 500)
company_Y <- c(380, 395, 410, 430, 460, 490)
company_Z <- c(450, 460, 440, 470, 510, 530)
month_num <- 1:6
plot(month_num, company_X,
     type = "o",
     col = "blue",
     pch = 16,
     xaxt = "n",
     ylim = c(350, 550),
     xlab = "Month",
     ylab = "Stock Price (₹)",
     main = "Monthly Stock Price Trend")
axis(1, at = month_num, labels = month)
lines(month_num, company_Y,
      type = "o",
      col = "red",
      pch = 17)
lines(month_num, company_Z,
      type = "o",
      col = "green",
      pch = 15)
legend("topleft",
       legend = c("Company X", "Company Y", "Company Z"),
       col = c("blue", "red", "green"),
       pch = c(16, 17, 15),
       lty = 1)
```

text(5, 520, "Highest X: ₹520", pos = 3)

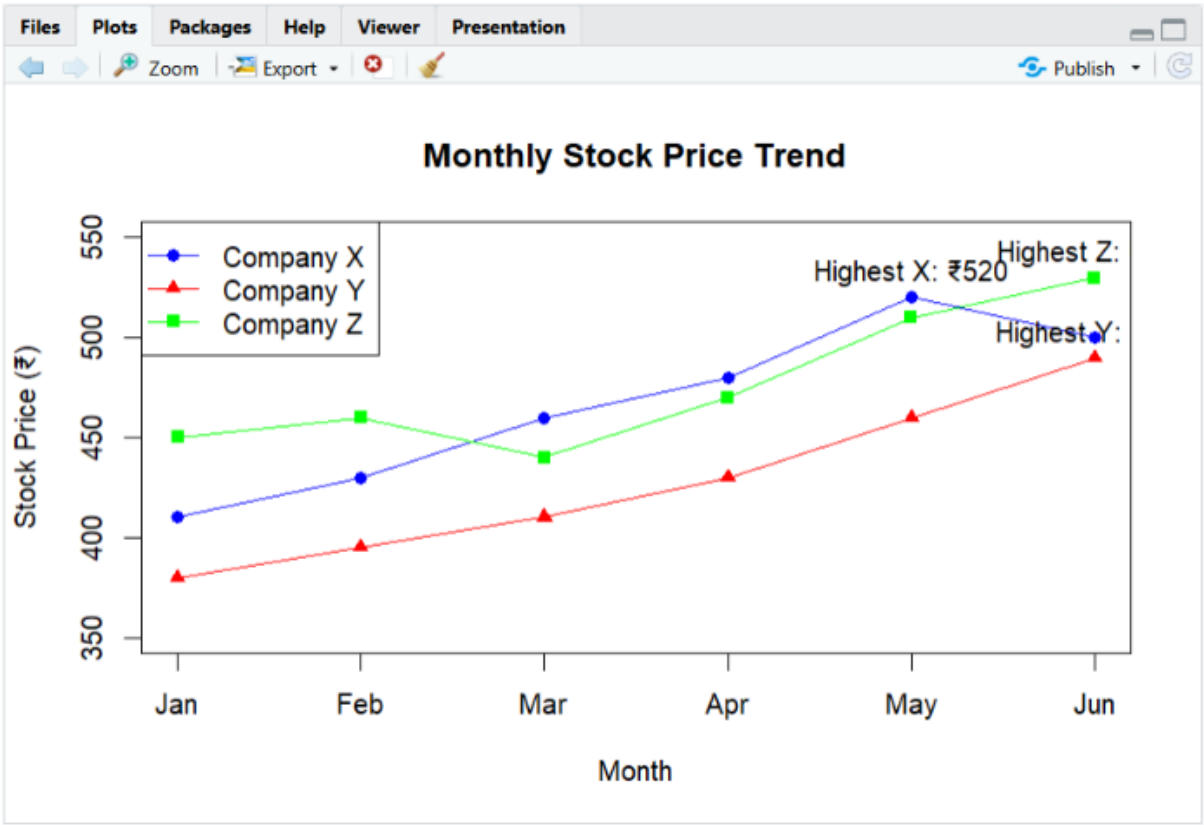
text(6, 490, "Highest Y: ₹490", pos = 3)

text(6, 530, "Highest Z: ₹530", pos = 3)

2. Annotate highest price

Highest values:

- Company X = **520** in May



- Company Y = **490** in June
- Company Z = **530** in June

Using text():

text("May", 520, labels = "Highest X: ₹520", pos = 3)

text("Jun", 490, labels = "Highest Y: ₹490", pos = 3)

text("Jun", 530, labels = "Highest Z: ₹530", pos = 3)

Add these lines **after the plot and lines commands**.

3. Role of legends

A legend explains which line, point, color, or symbol represents each category or company. It allows the viewer to correctly identify different data series.

4. Best practices for legends

- Place the legend where it does not cover important data.
- Use clear and short labels.
- Keep the legend close to the chart.
- Use consistent colors and symbols.
- Place it outside the plotting area when the chart is crowded.

5. Alternative to a legend

Direct labeling can be used. The company name can be written directly beside the final point of each line.

Example:

text("Jun", 500, "Company X", pos = 4)

text("Jun", 490, "Company Y", pos = 4)

text("Jun", 530, "Company Z", pos = 4)

Q3. Integrated Visualization — Python

For a bubble chart:

- **GDP** → X-axis
- **Literacy Rate** → Y-axis
- **Population** → Bubble size
- **Continent** → Bubble color

1 & 2. Python program

```
import matplotlib.pyplot as plt

country = ["C1", "C2", "C3", "C4", "C5"]

gdp = [12.5, 8.2, 21.0, 3.1, 5.6]

literacy = [92, 88, 95, 75, 80]

population = [140, 65, 330, 210, 95]

continent = ["Asia", "Europe", "N. America", "S. America", "Africa"]

colors = ["blue", "green", "red", "orange", "purple"]

plt.scatter(gdp, literacy,
```

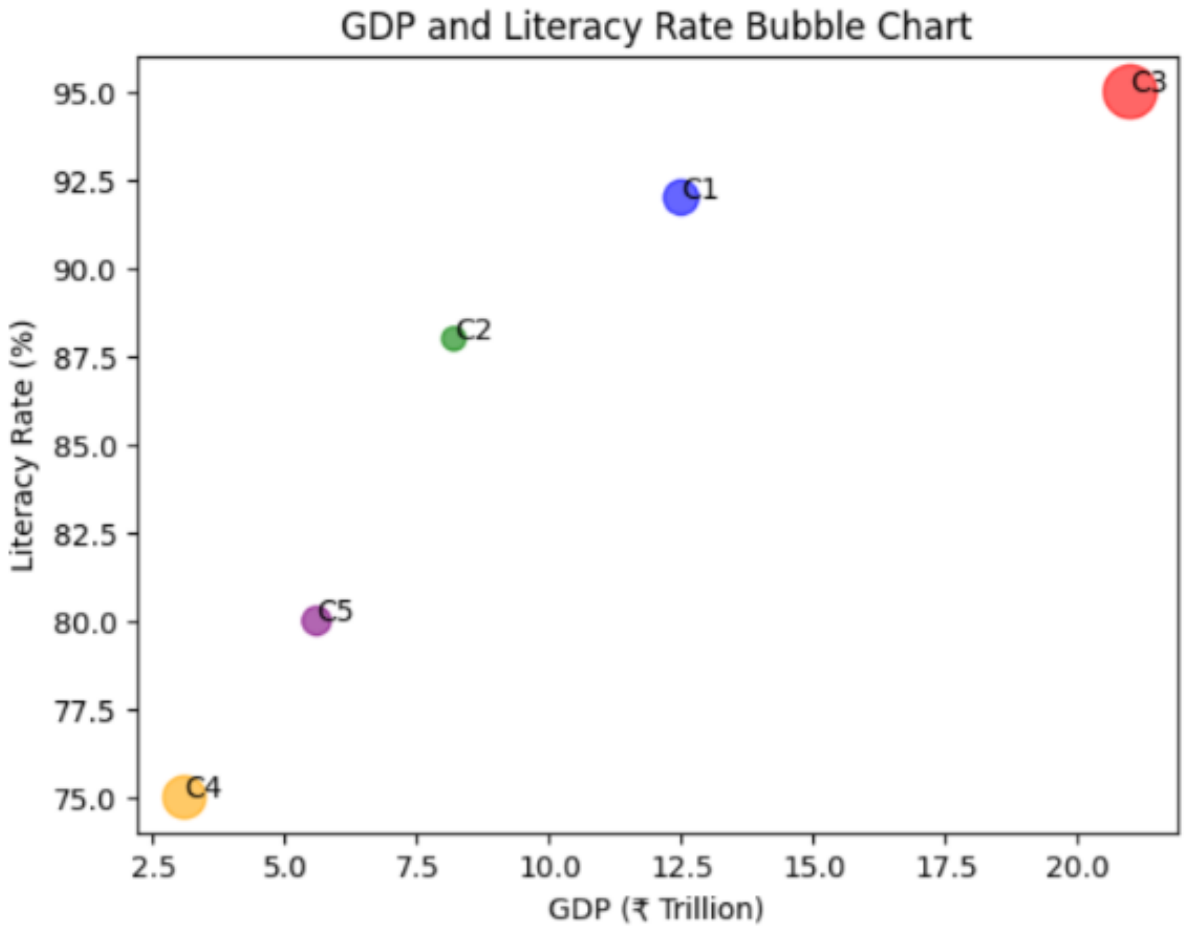
```
        s=population,
        c=colors,
        alpha=0.6)
for i in range(len(country)):
    plt.text(gdp[i], literacy[i], country[i])
plt.xlabel("GDP (₹ Trillion)")
plt.ylabel("Literacy Rate (%)")
plt.title("GDP and Literacy Rate Bubble Chart")
plt.show()
```

3. Relationship between GDP and Literacy Rate

From the given data, countries with higher GDP generally have higher literacy rates.

For example:

- C4: GDP = ₹3.1 trillion, Literacy = 75%



- C3: GDP = ₹21 trillion, Literacy = 95%

Thus, the data suggests a **positive relationship** between GDP and literacy rate. However, only five countries are given, so this should not be treated as proof of causation.

4. Continuous and categorical variables

Continuous variables:

- GDP
- Literacy Rate
- Population

Categorical variables:

- Country
- Continent

5. Improvement

One improvement is to use a **legend for continents** so that the colors can be easily understood.

Q4. Correlation Heatmap — R

1. R program

```
attendance <- c(92, 75, 88, 60, 95, 70)

study_hours <- c(3.5, 1.5, 2.8, 1.0, 4.0, 2.0)

sleep_hours <- c(6.5, 8.0, 7.0, 5.5, 7.5, 6.0)

marks <- c(88, 62, 80, 48, 91, 58)

data <- data.frame(

  Attendance = attendance,

  Study_Hours = study_hours,

  Sleep_Hours = sleep_hours,

  Marks = marks

)

cor_matrix <- cor(data)

print(cor_matrix)

heatmap(cor_matrix,

        main = "Correlation Heatmap",
```

col = colorRampPalette(c("blue", "white", "red"))(20),

symm = TRUE)

2. Strongest correlation

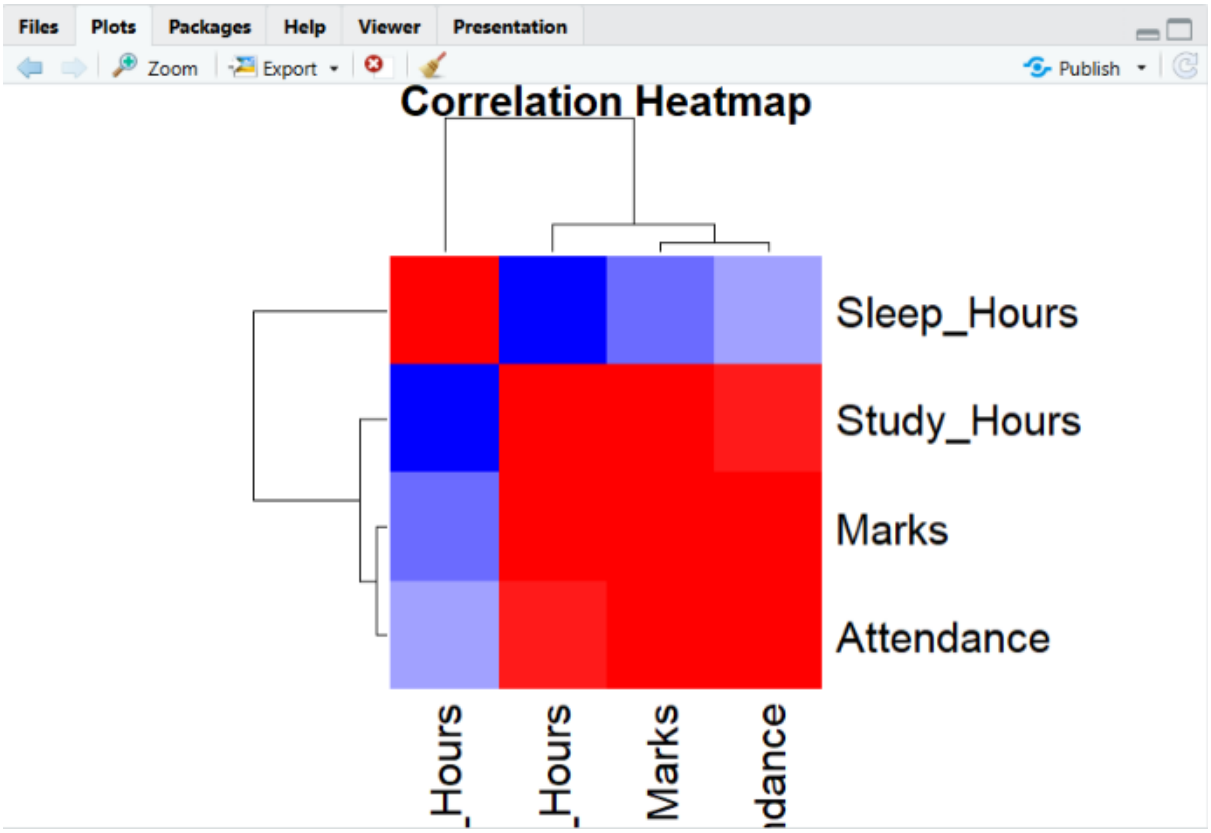
The strongest correlation is between:

Attendance and Marks

It is approximately **0.99**, indicating a very strong positive correlation.

Study Hours and Marks also show a strong positive relationship.

3.



Diverging color scale

A diverging color scale represents the **direction and strength of correlations**:

- One color → strong negative correlation
- White/neutral color → correlation near zero
- Opposite color → strong positive correlation

4. Limitation of correlation

Correlation shows an **association**, not causation.

For example, a high correlation between attendance and marks does not prove that attendance alone causes higher marks. Other factors may influence the results.

5. Additional variable

A useful additional variable would be:

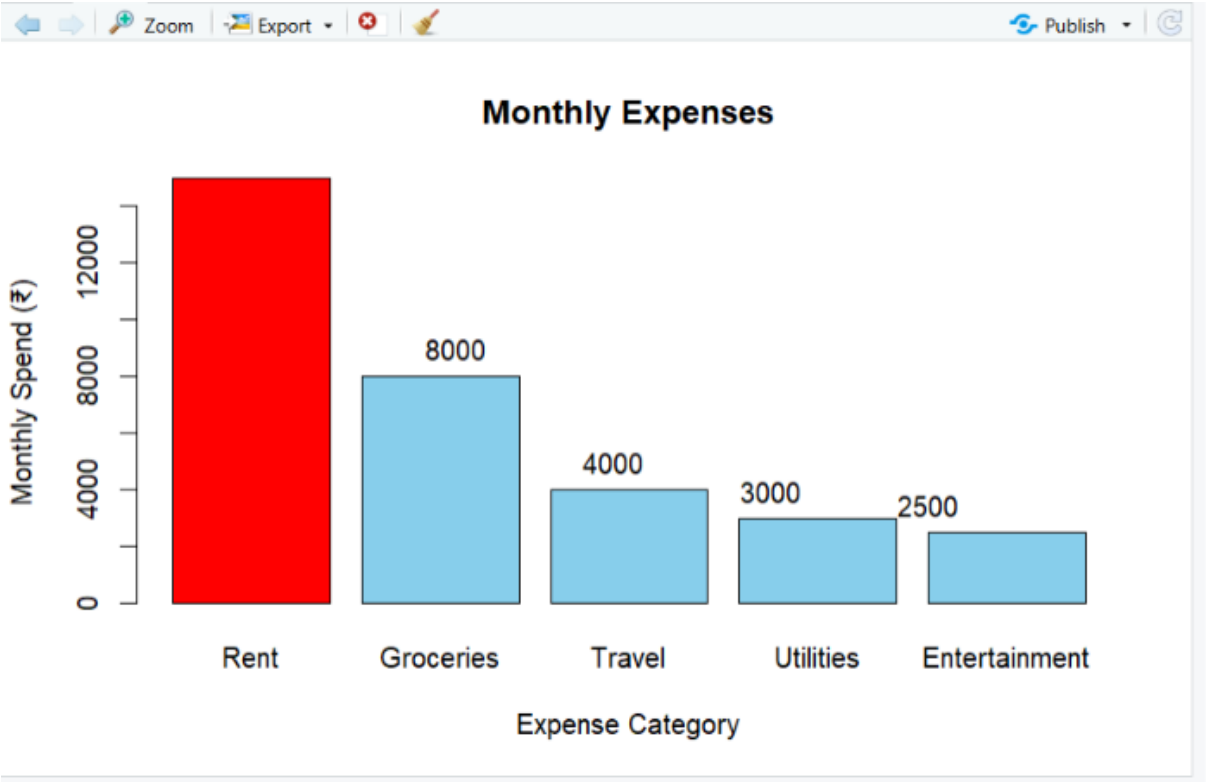
Previous academic performance / previous marks

Other possibilities include study method, class participation, or assignment scores.

Q5. Highlighting with Color — R

1. R program

```
category <- c("Rent", "Groceries", "Travel",
              "Utilities", "Entertainment")
spend <- c(15000, 8000, 4000, 3000, 2500)
highest <- which.max(spend)
bar_colors <- rep("skyblue", length(spend))
bar_colors[highest] <- "red"
barplot(spend,
        names.arg = category,
        col = bar_colors,
        main = "Monthly Expenses",
        xlab = "Expense Category",
        ylab = "Monthly Spend (₹)")
text(seq_along(spend),
     spend,
     labels = spend,
     pos = 3)
```



2. How highlighting improves interpretation

Highlighting draws attention to the most important data point. Here, using a different color for the highest expense makes **Rent** immediately noticeable.

3. Highest-spending category

Rent — ₹15,000

4. Effect of excessive colors

Using too many colors can make a chart:

- Confusing
- Difficult to read
- Visually distracting
- Harder to interpret

Colors should be used purposefully rather than simply for decoration.

5. Alternative highlighting technique

Instead of color, use **patterns, labels, arrows, or annotations**.

For example, an arrow can point to Rent and display:

"Highest: ₹15,000"